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Thermodynamic Modeling of the La-B and La-Bi Systems Supported by First-Principles Calculations

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Thermodynamic assessments of the La-B and La-Bi systems were carried out based on the available experimental data including thermodynamic properties and phase equilibria by means of the CALPHAD method. The enthalpies of formation at 0 K for the LaB_4, LaB_6, La_2Bi, La_5Bi_3, La_4Bi_3 and LaBi were computed via first-principles calculations. These enthalpies of formation were used "key experimental data" in the CALPHAD approach in order to obtain the thermodynamic parameters with sound physical meaning. The liquid phase has been modeled with both the substitutional solution model and associate model, and two sets of self-consistent thermodynamic parameters for the La-B and La-Bi systems were finally obtained. The calculated phase diagram and thermodynamic properties agree reasonably with the literature experimental data and the present first-principles calculations.

Keywords associate model, CALPHAD, first-principles calculations, lanthanum alloys, substitutional solution model

1. Introduction

The La-B and La-Bi binary systems have attracted attention due to the wide application of the alloys, which can be used as catalysts in the production of petroleum and synthetic products. The addition of lanthanum in Al alloys can improve the strength and workability of alloys. The design of La-B and La-Bi based novel materials requires information about the phase equilibria and thermodynamic properties in the La-B and La-Bi systems. However, to the best of our knowledge, no thermodynamic modeling has been performed for the La-B and La-Bi systems.

In this work, the available experimental data in the La-B and La-Bi systems have been critically assessed and used to evaluate the model parameters defining the Gibbs energies of the phases in these two systems. To supplement the experimental information, the enthalpies of formation (ΔH_f) for the intermetallic compounds in these two systems computed with first-principles method are also included. Finally, a set of self-consistent thermodynamic parameters is obtained for each of these two binary systems by means of CALPHAD approach.

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2. Literature Review

2.1 The La-B System

On the basis of thermal, metallographic, resistometric and X-ray analysis, Johnson and Daane[1] first proposed the La-B phase diagram. Two compounds LaB_4 and LaB_6 were reported. The homogeneity range of LaB_6 was determined to be from 85.5 to 86.5 at.% B.[1-3] Although similar homogeneity range for LaB_6 was detected by several researchers,[1-3] the lattice parameter of LaB_6 appears unaffected by B saturation. Therefore, LaB_6 was considered to be a line compound in the present assessment. Subsequently, a new stoichiometric compound LaB_9 was found by Gert et al.,[4] which was confirmed by Storms and Mueller.[2] The decomposition temperature of LaB_9 was detected to be at 2007 °C by using the thermal analysis method. This phase was considered in the present work, although the crystal structure and lattice parameter of LaB_9 are still unknown.

The thermodynamic properties of the La-B system reported in the literature are limited, and most published data refer to the LaB_4 and LaB_6 phases.[5-10] The heat contents for LaB_6 were measured by Westrum et al.,[5] Kaprina et al.[7] and Bolgar et al.[8] The enthalpy of formation of the LaB_6 phase at 25 °C was determined by Topor and Kleppa[9] using the solute-solvent drop calorimetry measurement. Besides, the enthalpy of formation of LaB_4 was measured by Meschel and Kleppa[10] using the direct synthesis calorimetry method. Using the vapor pressure method, the activities of B and La were determined by Storms and Mueller[2] at 1627 °C. All the experimental thermodynamic data were considered in the present assessment. The experimental phase diagram data and thermodynamic properties have been summarized by Schlesinger.[1]

2.2 The La-Bi System

The La-Bi phase diagram was determined by several groups of authors.[12-15] Five intermetallic compounds

Table 1 Summary of the experimental, estimated phase diagram and thermodynamic data in the La-B and La-Bi systems

System	Type of experimental data	Method	Quoted mode(a)	Ref.
La-B	Phase relationship from 0 to 50 at.% B	TA	+	[1]
	Phase relationship from 80 to 100 at.% B	TA	+	[2]
	Activity of La and B in the solid phases at 1627 °C	VPM	■	[2]
	Heat contents for LaB_6 from 120 to 730 °C	…	+	[5]
	Heat capacities and enthalpy of formation for LaB_6	…	■	[6]
	Heat contents for LaB_6 from 820 to 1950 °C	HTC	+	[7]
	Heat contents for LaB_6 from 190 to 1938 °C	…	■	[8]
	Enthalpy of formation for LaB_6	SSDC	+	[9]
	Enthalpy of formation for LaB_4	DSC	■	[10]
La-Bi	The phase diagram of whole composition range	DTA	+	[14]
	The solubilities of La in Bi	…	■	[12,13]
	Heat capacities of LaBi from 7 to 300 K	AC	□	[22]
	The free Gibbs energy for La_2Bi from 550 to 750 °C	emf	■	[21]
	Activity coefficient of La in the Bi-rich side	emf	■	[13,16-18]
	Activity of Bi in the solid phases at 627°C	VPM	□	[19]
	Enthalpy of formation for LaBi and La_4Bi_3	DIC	+	[20]
	Enthalpy of formation for LaBi	Estimated	■	[23]
	Enthalpy of formation for La_2Bi, La_4Bi_3, La_5Bi_3, LaBi, $LaBi_2$	Estimated	■	[24]
	The free Gibbs energy of LaBi from 0 to 1527°C	Estimated	□	[25]
	The free Gibbs energy of LaBi2 at 600 °C	Estimated	□	[26]

TA, thermal analysis; DTA, differential thermal analysis; SSDC, solute-solvent drop calorimetry; DSC, direct synthesis calorimetry; AC, adiabatic calorimeter; DIC, direct isoperibol calorimeter; VPM, vapor pressure method; HTC, high temperature calorimeter; emf, electromotive force

(a) Indicates whether the data are used or not used in the parameter optimization: + used; ■ not used but estimated to be reliable data for checking the modeling; □ not used

$LaBi_2$, LaBi, La_4Bi_3, La_5Bi_3 and La_2Bi were reported by Nomura et al.[14] using X-ray diffraction (XRD) analysis and microscopic examination. Except for the La_4Bi_3 phase, which melts congruently at 1670 °C, the other four compounds are formed via peritectic reactions. The transformation temperatures were measured by means of the differential thermal analysis (DTA). The solubilities of La in Bi were determined by Schweitzer and Weeks[12] and Kober et al.[13] All the reported experimental data were considered in the present assessment.

Several thermodynamic properties for the La-Bi system were reported in the literature.[13,16-22] By means of the electromotive force (emf) method, the activity coefficient of La in the Bi-rich side was determined.[13,16-18] The Gibbs free energy of $LaBi_2$ from 550 to 750 °C was determined by Morisson and Petot[21] using the emf method. The activity of Bi at 627 °C was measured by Shoji et al.[19] using the vapor pressure method. The enthalpies of formation for the LaBi and La_4Bi_3 phases were determined by Borsese et al.[20] by means of direct isoperibol calorimeter. The heat capacities of LaBi phase at low temperatures (from 7 to 298 K) were measured by Wallace et al.,[22] only the data at 298 K was considered since the present work only considers the data above the room temperature. As the reported activity coefficients are rather small, they are not used during the present assessment, but employed to check the reliability of the present calculation. Besides, the enthalpies of formation and Gibbs free energies of the compounds were estimated by several groups of authors,[23-26] which were not used in the present assessment since these data are too scattering. All the experimental and estimated data are summarized in Table 1. The La-Bi system has been reviewed by several groups of researchers,[27-31] who considered the experimental data before the year 1996.

3. First-Principles Calculations

The existing thermochemical information for the intermetallic compounds available in the literature is rather uncertain, as the scatters between the measured data were considerable. Thus, first-principles calculations were employed to test the consistency of the experimental data. Besides, this approach allowed the characterization of the thermochemical properties of La_2Bi and La_5Bi_3 phases, the experimental data of which are not available in the literature.

In the present work, the enthalpies of formation for LaB_4, LaB_6, La_2Bi, La_5Bi_3, La_4Bi_3 and LaBi were obtained using first-principles method. The total energies of LaB_4, LaB_6, La_2Bi, La_5Bi_3, La_4Bi_3, LaBi, La, B, and Bi were calculated

Table 2 Crystal structure data for the phases in the La-B and La-Bi systems according to the present first-principles calculations at 0 K along with the literature data

Phase	Pearson symbol	Protype	Space group	Lattice parameters, nm		
				Exp.[45]	Cal.	Diff. (%)
Bcc(La)	*cI*2	W	$Im\bar{3}m$	a = 0.4290	a = 0.4250	−0.93
Fcc(La)	*cF*4	Cu	$Fm\bar{3}m$	a = 0.5290	a = 0.5260	−0.59
Dhcp(La)	*hP*4	(αLa)	$P6_3/mcm$	a = 0.3772	a = 0.3762	−0.27
				c = 1.2144	c = 1.2078	−0.54
(αB)	*hR*12	(αB)	$R\bar{3}m$	a = 0.5057	a = 0.5052	−0.10
(βB)	*hR*108	(βB)	$R\bar{3}m$	a = 1.017	a = 1.012	−0.49
(Bi)	*hR*2	As	$R\bar{3}m$	a = 0.4546	a = 0.4583	0.81
LaB_4	*tP*20	B4Th	$P4/mbm$	a = 0.7325	a = 0.7310	−0.20
				c = 0.4181	c = 0.4184	0.07
LaB_9	…	…	…	…	…	…
LaB_6	*cP*7	B_6Ca	$Pm\bar{3}m$	a = 0.4151	a = 0.4155	0.10
La_2Bi	*tI*12	La_2Sb	$I4/mmm$	a = 0.4737	a = 0.4712	−0.53
				c = 1.8340	c = 1.8583	1.32
La_5Bi_3	*hP*16	Mn_5Si_3	$P6_3/mcm$	a = 0.9659	a = 0.9767	0.81
				c = 0.6697	c = 0.6727	0.45
La_4Bi_3	*cI*28	P_4Th_3	$I\bar{4}3d$	a = 0.9790	a = 0.9886	0.98
LaBi	*cF*8	NaCl	$Fm\bar{3}m$	a = 0.6658	a = 0.6659	0.02
$LaBi_2$	*oC*12	$LaSb_2$	$I4/mmm$	a = 0.4737	…	…
				b = 1.751	…	…
				c = 0.4564	…	…

using density functional theory (DFT)[32] within a generalized gradient approximation (GGA)[33] with projector augmented-wave (PAW) pseudopotentials,[34] as implemented in the Vienna ab initio simulation package (VASP).[35,36] The GGA proposed by Perdew, Burke and Ernzerhof (PEB)[37] was used in the calculation. Brillouin zone integrations were performed using a Monkhorst-Pack mesh.[38] 400 eV energy cut off was used in the present work, and the total energy was converged to be less than 1 meV/atom. The structure was fully relaxed using the Methfessel-Paxton smearing method,[39] and a final self-consistent static calculation via the tetrahedron smearing method with Blöchl corrections[40] was performed.

The enthalpy of formation of a compound is defined as difference between the energy of the compound and the sum of energies of its constituent elements in their stable states:

$$\Delta H_f = E_{total}^{La_xM_{1-x}} - xE_{total}^{La-DHcp} - (1-x)E_{total}^{M} \quad \text{(Eq 1)}$$

where $E_{total}^{La_xM_{1-x}}$ is the total energy of compound La_xM_{1-x}, and $E_{total}^{La-DHcp}$ and E_{total}^{M} are the total energies of La and M (B and Bi) in their stable structures, respectively. Since the influence of pressure on the condensed phases was ignored and the energy was calculated at 0 K without any entropic contributions, the calculated energy of formation was taken to be the enthalpy of formation at 298 K. The first-principles calculation is not employed to calculate the enthalpy of formation for the compounds LaB_9 and $LaBi_2$ for the Wykoff position of atoms in LaB_9 and $LaBi_2$ were not determined.

4. Thermodynamic Modeling

4.1 Unary Phases

The Gibbs energy function ${}^{\circ}G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER}$ for the pure element i (i = La, B and Bi) in the φ phase is described by the following equation:

$$ {}^{\circ}G_i^{\varphi}(T) = a + b \cdot T + c \cdot T \cdot \ln(T) + d \cdot T^2 + e \cdot T^3 + f \cdot T^{-1} + g \cdot T^7 + h \cdot T^{-9} \quad \text{(Eq 2)}$$

where H_i^{SER} is the molar enthalpy of element i at 298.15 K and 1 bar in its standard element reference (SER). T is the absolute temperature. In the present work, the Gibbs energies for La, B and Bi are taken from the SGTE compilation by Dinsdale.[41]

4.2 Stoichiometric Compounds

In view of the negligible homogeneities for binary compounds in the La-B and La-Bi systems, all the compounds are treated as stoichiometric phases. The Gibbs energy of $La_xM_{(1-x)}$ (M = B, Bi) per mole atom is given by the following expression:

$$ {}^{\circ}G_m^{La_xM_{(1-x)}} = x \cdot {}^{\circ}G_{La}^{DHcp} + (1-x) \cdot {}^{\circ}G_M + A + B \cdot T \quad \text{(Eq 3)}$$

The parameters ${}^{\circ}G_{La}^{DHcp}$ and ${}^{\circ}G_M$ are the Gibbs energies of pure La and M (M = B, Bi) in the stable structure, respectively.

Table 3 The optimized thermodynamic parameters of the La-B and La-Bi systems using both the substitutional solution model and associate model

Systems	Phases	Set 1	Set 2
La-B	Liquid	Substitutional solution model: (La, B) ${}^{0}L^{\mathrm{Liquid}}_{\mathrm{La,B}} = 57705.6 - 111.7581 \cdot T$ ${}^{1}L^{\mathrm{Liquid}}_{\mathrm{La,B}} = -259928.9 + 82.7573 \cdot T$ ${}^{2}L^{\mathrm{Liquid}}_{\mathrm{La,B}} = -95244.5$	Associate model (La, LaB_6, B) $G^{\mathrm{Liquid}}_{\mathrm{LaB_6}} = -55949.0 - 86.7600 \cdot T + {}^{\circ}G^{\mathrm{Liquid}}_{\mathrm{La}} + 6 \cdot {}^{\circ}G^{\mathrm{Liquid}}_{\mathrm{B}}$ ${}^{\circ}L^{\mathrm{Liquid}}_{\mathrm{La,LaB_6}} = -295285.3$ ${}^{\circ}L^{\mathrm{Liquid}}_{\mathrm{B,LaB_6}} = -187858.0 + 39.0826 \cdot T$
	LaB_4 $(La)_{1/5}(B)_{4/5}$	$G^{\mathrm{LaB_4}}_{\mathrm{La:B}} = -59937.8 + \frac{1}{5} \cdot {}^{\circ}G^{\mathrm{DHcp}}_{\mathrm{La}} + \frac{4}{5} \cdot {}^{\circ}G^{\mathrm{Beta}}_{\mathrm{B}}$	$G^{\mathrm{LaB_4}}_{\mathrm{La:B}} = -58451.4 + \frac{1}{5} \cdot {}^{\circ}G^{\mathrm{DHcp}}_{\mathrm{La}} + \frac{4}{5} \cdot {}^{\circ}G^{\mathrm{Beta}}_{\mathrm{B}}$
	LaB_6 $(La)_{1/7}(B)_{6/7}$	$G^{\mathrm{LaB_6}}_{\mathrm{La:B}} = -66034.1 + 125.4638 \cdot T - 19.6431 \cdot T \cdot \ln(T) - 0.0043907 \cdot T^2 + 355709.6 \cdot T^{-1}$	$G^{\mathrm{LaB_6}}_{\mathrm{La:B}} = -66832.8 + 125.5823 \cdot T - 19.64305 \cdot T \cdot \ln(T) - 0.00439067 \cdot T^2 + 355709.64 \cdot T^{-1}$
	LaB_9 $(La)_{1/10}(B)_{9/10}$	$G^{\mathrm{LaB_9}}_{\mathrm{La:B}} = -42761.1 + \frac{1}{10} \cdot {}^{\circ}G^{\mathrm{DHcp}}_{\mathrm{La}} + \frac{9}{10} \cdot {}^{\circ}G^{\mathrm{Beta}}_{\mathrm{B}}$	$G^{\mathrm{LaB_9}}_{\mathrm{La:B}} = -43121.1 + \frac{1}{10} \cdot {}^{\circ}G^{\mathrm{DHcp}}_{\mathrm{La}} + \frac{9}{10} \cdot {}^{\circ}G^{\mathrm{Beta}}_{\mathrm{B}}$
La-Bi	Liquid	Substitutional solution model: (La, Bi) ${}^{0}L^{\mathrm{Liquid}}_{\mathrm{La,Bi}} = -262644.9 + 2.8121 \cdot T$ ${}^{1}L^{\mathrm{Liquid}}_{\mathrm{La,Bi}} = -4647.4 - 9.5176 \cdot T$ ${}^{2}L^{\mathrm{Liquid}}_{\mathrm{La,Bi}} = 61044.5$	Associate model (La, La_4Bi_3, Bi) $G^{\mathrm{Liquid}}_{\mathrm{La_4Bi_3}} = -180006.4 - 75.2523 \cdot T + 4 \cdot {}^{0}G^{\mathrm{Liquid}}_{\mathrm{La}} + 3 \cdot {}^{0}G^{\mathrm{Liquid}}_{\mathrm{Bi}}$ ${}^{0}L^{\mathrm{Liquid}}_{\mathrm{La,La_4Bi_3}} = -389753.4 + 23.5975 \cdot T$ ${}^{0}L^{\mathrm{Liquid}}_{\mathrm{Bi,La_4Bi_3}} = -646034.0 + 92.8484 \cdot T$
	Bcc_A2 $(La, Bi)_1Va_3$	${}^{0}L^{\mathrm{Bcc_A2}}_{\mathrm{La,Bi:Va}} = -266505.2 + 69.5274 \cdot T$	${}^{0}L^{\mathrm{Bcc_A2}}_{\mathrm{La,Bi:Va}} = -275208.3 + 78.7447 \cdot T$
	La_2Bi $(La)_{2/3}(Bi)_{1/3}$	$G^{\mathrm{La_2Bi}}_{\mathrm{La:Bi}} = -74397.6 + \frac{1}{3} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{2}{3} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$	$G^{\mathrm{La_2Bi}}_{\mathrm{La:Bi}} = -74156.8 + \frac{1}{3} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{2}{3} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$
	La_5Bi_3 $(La)_{5/8}(Bi)_{3/8}$	$G^{\mathrm{La_5Bi_3}}_{\mathrm{La:Bi}} = -83404.5 + 0.8973 \cdot T + \frac{3}{8} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{5}{8} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$	$G^{\mathrm{La_5Bi_3}}_{\mathrm{La:Bi}} = -81116.3 + \frac{3}{8} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{5}{8} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$
	La_4Bi_3 $(La)_{4/7}(Bi)_{3/7}$	$G^{\mathrm{La_4Bi_3}}_{\mathrm{La:Bi}} = -95066.6 + 2.4855 \cdot T + \frac{3}{7} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{4}{7} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$	$G^{\mathrm{La_4Bi_3}}_{\mathrm{La:Bi}} = -90184.0 + 0.7093 \cdot T + \frac{3}{7} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{4}{7} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$
	LaBi $(La)_{1/2}(Bi)_{1/2}$	$G^{\mathrm{LaBi}}_{\mathrm{La:Bi}} = -111734.3 + 10.2722 \cdot T + \frac{1}{2} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{1}{2} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$	$G^{\mathrm{LaBi}}_{\mathrm{La:Bi}} = -103120.9 + 5.9546 \cdot T + \frac{1}{2} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{1}{2} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$
	$LaBi_2$ $(La)_{1/3}(Bi)_{2/3}$	$G^{\mathrm{LaBi_2}}_{\mathrm{La:Bi}} = -85812.6 + 8.5518 \cdot T + \frac{2}{3} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{1}{3} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$	$G^{\mathrm{LaBi_2}}_{\mathrm{La:Bi}} = -72824.0 + 0.7611 \cdot T + \frac{2}{3} \cdot {}^{\circ}G^{\mathrm{Rhombo}}_{\mathrm{Bi}} + \frac{1}{3} \cdot {}^{\circ}G^{\mathrm{Dhcp}}_{\mathrm{La}}$

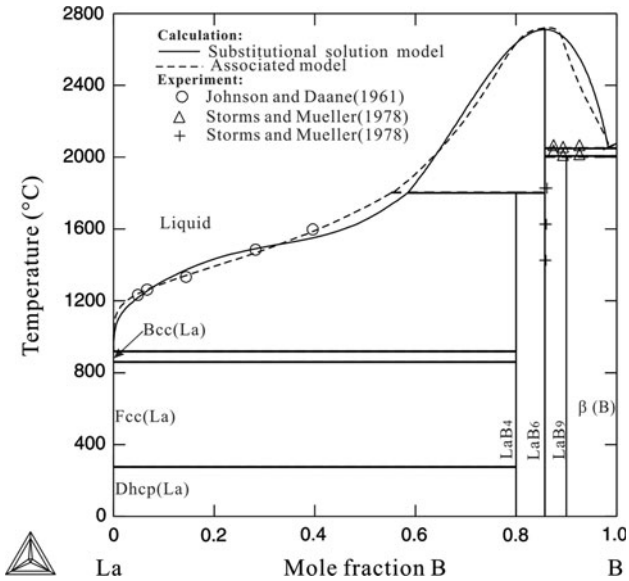

Fig. 1 Calculated La-B phase diagram with both substitutional solution model (solid line) and associate model (dashed line), compared with the experimental data[1,2]

Since the heat contents of LaB_6 were measured between the temperature of 120 and 1950 °C by Westrum et al.,[5] Kapyina et al.[7] and Bolgar et al.,[8] it is preferable to express its Gibbs energy relative to the SER state:

$$^{\circ}G_m^{LaB_6} = 1/7 \cdot {}^{\circ}H_{La}^{DHcp} + 6/7 \cdot {}^{\circ}H_B^{Beta} + A + B \cdot T + C \cdot T \cdot \ln(T) + D \cdot T^2 + E \cdot T^{-1} + F \cdot T^3 \quad \text{(Eq 4)}$$

The coefficients C, D, E and F can be directly determined from the measured heat contents.

4.3 Solid Solution Phases

The solid solution phases Fcc_A1 and Bcc_A2 are described by Redlich-Kister polynomial.[42] The Gibbs energy is expressed as follows:

$$^{\circ}G_m^{\varphi} = x_1 \cdot {}^{\circ}G_1 + x_2 \cdot {}^{\circ}G_2 + R \cdot T \cdot (x_1 \cdot \ln x_1 + x_2 \cdot \ln x_2) + x_1 \cdot x_2 \cdot \sum_{v=0}^{n} {}^{v}L \cdot (x_1 - x_2)^v \quad \text{(Eq 5)}$$

where R is the gas constant, and x is the mole fraction of component. L in the third term is the interaction parameter between the elements, which is expressed as follows:

$$^{v}L = a_v + b_v \cdot T \quad \text{(Eq 6)}$$

The parameters of a_v and b_v are to be evaluated based on the experimental data.

4.4 The Liquid Phase

The relatively high melting points of the LaB_6 and La_4Bi_3 phases suggest the existence of a short-range order in the liquid phase. Thus by assuming the existence of short-lived associates of LaB_6 and La_4Bi_3, the associate model in the form of (La, LaB_6, B) and (La, La_4Bi_3, Bi) is used to describe the liquid phase in the La-B and La-Bi systems, respectively. Meanwhile, the liquid phase of the La-B and La-Bi systems is also described using the substitutional solution model in the form of (La, B) and (La, Bi) in order to be compatible with multi-component databases, in which the liquid phase is generally described with substitutional solution model. With the changing of the model for the liquid phase, the phase equilibria for most of the phases in these two systems will be affected to a certain extent. Therefore, in the present work "set 1" and "set 2" in Table 1 mean the two sets of thermodynamic parameters obtained with the liquid phases assessed with substitutional solution model and associate model, respectively.

The molar Gibbs energy of substitutional solution model for liquid phase is given with the same form as that for the solid solution phase (Eq 5 and 6). The molar Gibbs energy of liquid described with the associate model is given by an equation of the form:

$$^{\circ}G_m^{Liquid} - x_{La} \cdot H_{La}^{SER} - x_M \cdot H_M^{SER} = \sum_{i=La,M,N} x_i \cdot {}^{\circ}G_i^{Liquid} + R \cdot T \cdot \sum_{i=La,M,N} x_i \cdot \ln x_i + {}^{ex}G \quad \text{(Eq 7)}$$

where M = B or Bi and N = LaB_6 or La_4Bi_3. The Gibbs energy of LaB_6 associate is given by:

$$^{\circ}G_{LaB_6}^{L} = a + b \cdot T + {}^{\circ}G_{La}^{L} + 6 \cdot {}^{\circ}G_B^{L} \quad \text{(Eq 8)}$$

The parameters a and b are to be evaluated. The Gibbs energy for La_4Bi_3 associate is analogous to that of the LaB_6. Due to the associate's stoichiometry, the excess Gibbs energy of the liquid phase can be described by the interactions between La and LaB_6 (La_4Bi_3), between B (Bi) and LaB_6 (La_4Bi_3) as well as between La and B (Bi), resulting in the following expression:

$$^{ex}G = x_{La} \cdot x_N \cdot L_{La,N}^{Liquid} \cdot (x_N^{Liquid} - x_{La}^{Liquid}) + x_{La} \cdot x_M \cdot L_{La,M}^{Liquid} \cdot (x_{La}^{Liquid} - x_M^{Liquid}) + x_M \cdot x_N \cdot L_{M,N}^{Liquid} \cdot (x_N^{Liquid} - x_M^{Liquid}) \quad \text{(Eq 9)}$$

5. Results and Discussion

The evaluation of thermodynamic parameters were carried out by using the PARROT module in the Thermo-Calc software,[43] which works by minimizing the squared sum of differences between experimental values and computed ones. The step by step optimization procedure carefully described by Du et al.[44] was utilized in the present work. In the optimization, each piece of experimental information was given a certain weight based on uncertainties of the data. Reliable experimental information including phase diagram[1,2,14] and thermodynamic

Table 4 Calculated invariant reactions using both the substitutional solution model and associate model compared with the experimental data of the La-B and La-Bi systems

Systems	Reactions	Compositions			*T*, °C	Ref.
La-B		at.% B				
	Liquid = LaB_6	85.714	…	…	2715	[2] (Exp.)
		85.714	…	…	2710.1	This work (Cal. set 1)
		85.714	…	…	2714.8	This work (Cal. set 2)
	Liquid = LaB_6 + (B)	95.0	85.714	…	2052	[2] (Exp.)
		98.38	85.714	…	2049.5	This work (Cal. set 1)
		98.00	85.714	100.0	2051.9	This work (Cal. set 2)
	LaB_6 + (B) = LaB_9	85.714	…	90.0	2007	[2] (Exp.)
		85.714	100.0	90.0	2005.7	This work (Cal. set 1)
		85.714	100.0	90.0	2002.9	This work (Cal. set 2)
	Liquid + LaB_6 = LaB_4	66.7	85.714	80.0	1800	[2] (Exp.)
		58.47	85.714	80.0	1801.8	This work (Cal. set 1)
		55.19	85.714	80.0	1802.8	This work (Cal. set 2)
	Liquid = LaB_4 +Bcc(La)	0.1	…	80.0	918	[1] (Exp.)
		0.02	0	80.0	919.7	This work (Cal. set 1)
		0.01	0	80.0	919.9	This work (Cal. set 2)
La-Bi		at.% Bi				
	Bcc(La) = La_2Bi +Fcc(La)	…	…	33.33	750.0	[14] (Exp.)
		0	2.18	33.33	749.4	This work (Cal. set 1)
		0	2.18	33.33	749.3	This work (Cal. set 2)
	Liquid = Bcc(La) + La_2Bi	…	…	33.33	820.0	[14] (Exp.)
		2.05	8.68	33.33	825.1	This work (Cal. set 1)
		1.97	8.66	33.33	824.9	This work (Cal. set 2)
	Liquid + La_5Bi_3 = La_2Bi	…	33.33	37.50	1252 ± 20	[14] (Exp.)
		23.62	33.33	37.50	1249.2	This work (Cal. set 1)
		26.15	33.33	37.5	1251.6	This work (Cal. set 2)
	Liquid + La_4Bi_3 = La_5Bi_3		37.50	42.86	1350 ± 20	[14] (Exp.)
		26.92	37.50	42.86	1346.3	This work (Cal. set 1)
		28.54	37.50	42.86	1347.3	This work (Cal. set 2)
	Liquid = La_4Bi_3	42.86	…	…	1670 ± 30	[14] (Exp.)
		42.86	…	…	1670.4	This work (Cal. set 1)
		42.86	…	…	1678.3	This work (Cal. set 2)
	Liquid + La_4Bi_3 = LaBi	…	42.86	50.00	1615 ± 15	[14] (Exp.)
		50.07	42.86	50.00	1588.7	This work (Cal. set 1)
		50.45	42.86	50.00	1611.7	This work (Cal. set 2)
	Liquid + LaBi = $LaBi_2$	…	50.00	66.67	932 ± 10	[14] (Exp.)
		75.35	50.00	66.67	937.0	This work (Cal. set 1)
		71.10	50.00	66.67	936.8	This work (Cal. set 2)
	Liquid = $LaBi_2$ + (Bi)	…	66.67	…	271.4	[14] (Exp.)
		99.97	66.67	100.00	271.4	This work (Cal. set 1)
		99.98	66.67	100.00	271.1	This work (Cal. set 2)

data[5,7,9,14,20] in the La-B and La-Bi systems were employed in the optimization. Besides, as shown in Table 2, the differences between the experimental and calculated lattice parameters are insignificant (the maximum deviation is 1.32%) during the first-principles calculations. It indicates that the present calculation is reliable. Therefore, the enthalpies of formation computed by first-principles were also taken into account in the present assessment.

Taking the La-Bi system as example, the optimization began with La_4Bi_3 phase. The enthalpy formation of La_4Bi_3 calculated via the first-principles method were used as good starting values for the parameter A in Eq 3. Then, the parameters a_0 and b_0 in Eq 6 for liquid phase were adjusted to reproduce experimental liquidus data associated with La_4Bi_3 phase. The value of B in Eq 3 was also obtained by using phase diagram data associated with liquid phase and La_4Bi_3. Subsequently, La_5Bi_3, LaBi, $LaBi_2$ and La_2Bi were introduced in the modeling one by one. Finally, the thermodynamic parameters for all phases were adjusted simultaneously by taking into account all of selected phase equilibrium and thermodynamic data. It was found that introducing a_1, b_1 and a_2 for liquid phase in Eq 6 could yield a significant improvement to the description of liquidus. The optimization procedure for the La-B binary

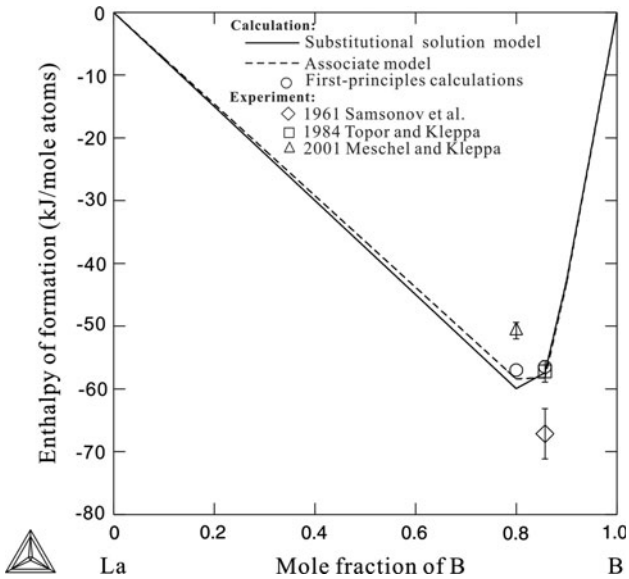

Fig. 2 Calculated enthalpy of formation at 298.15 K of the La-B system with both substitutional solution model (solid line) and associate model (dashed line), compared with the experimental data[6,9,10] and first-principles calculations

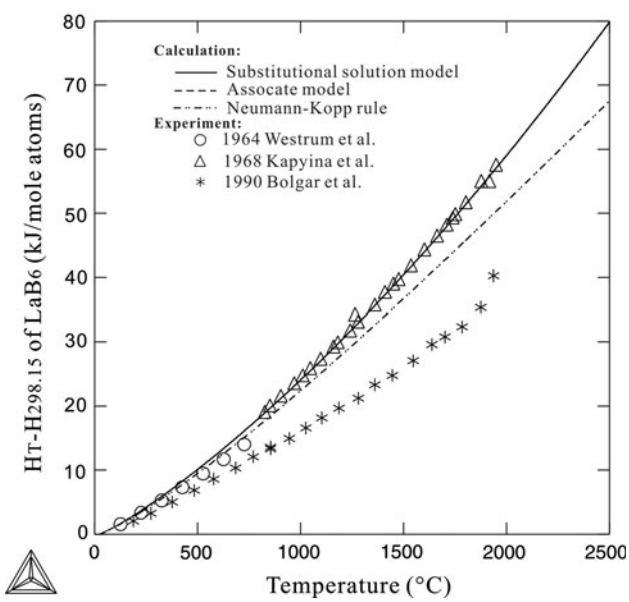

Fig. 3 Calculated heat contents of the LaB_6 phase in the La-B system compared with the experimental data[5,7,8]

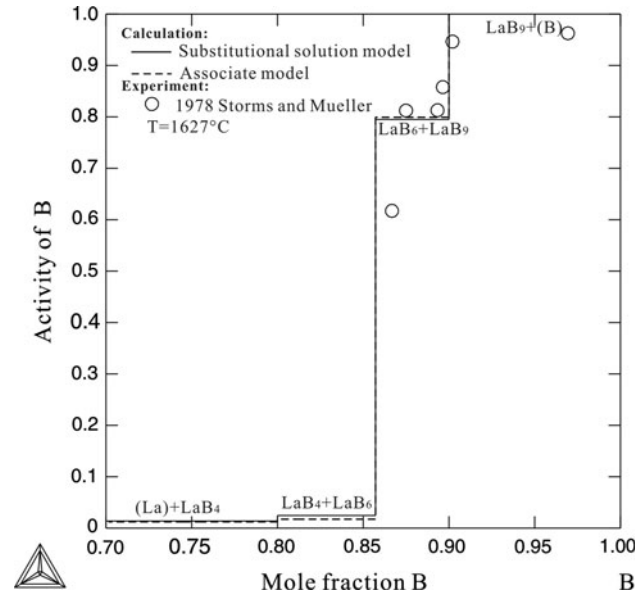

Fig. 4 Calculated activity of B at 1627 °C in the La-B system compared with the experimental data[2] in the solid phases

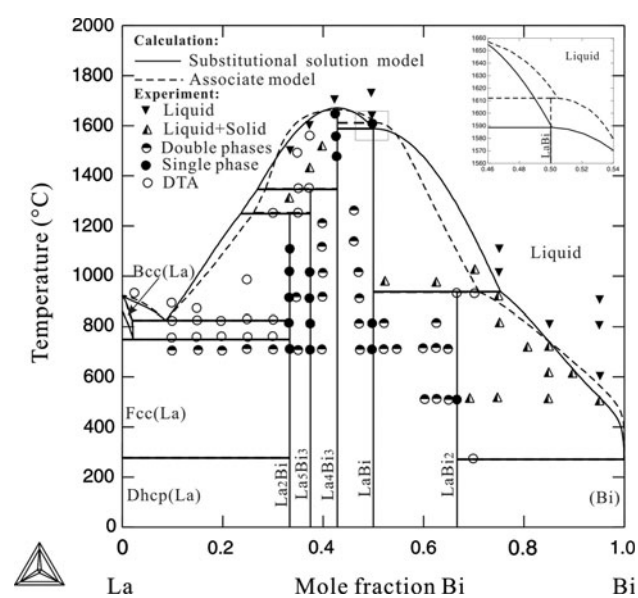

Fig. 5 Calculated La-Bi phase diagram with both substitutional solution model (solid line) and associate model (dashed line), compared with the experimental data[14]

system is analogous to that of the La-Bi system. As for the LaB_6 phase, the parameters C, D, E and F in Eq 4 were determined from the H_T-$H_{298.15}$ experimental data. The finally obtained parameters for the La-B and La-Bi binary systems are listed in Table 3.

5.1 The La-B System

Figure 1 shows the La-B phase diagram calculated using two sets of the present optimized parameters (solid line for set 1 and dashed line for set 2) in comparison with the available experimental data.[1,2] This figure shows that the present calculation using both the associate and substitutional solution models can reproduce the experimental data well. Table 4 presents the summary of the temperatures and compositions for the invariant reactions in the La-B binary system. Table 4 shows that the maximum difference between the calculated and the measured temperatures is within the experimental errors.

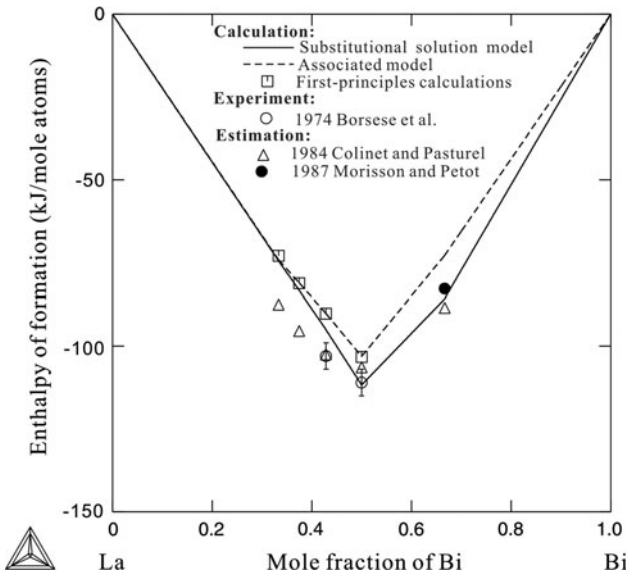

Fig. 6 Calculated enthalpy of formation at 298.15 K in the La-Bi system with both substitutional solution model (solid line) and associate model (dashed line), compared with the experimental data,[20] estimated data[23,24] and first-principles calculations

The calculated enthalpies of formation at 298.15 K from CALPHAD modeling, compared with those from first-principles calculations and experimental data,[9,10] are shown in Fig. 2. The agreement demonstrates that first-principles calculations can provide reliable values when the experimental data are not available. A further check on the reliability of the thermodynamic modeling is provided by Fig. 3, in which the heat capacities of compound LaB_6 were computed in the present modeling, compared with the experimental data from Westrum et al.,[5] Kapyina et al.[7] and Bolgar et al.[8] There is a reasonable agreement between the calculated values and the experimental data. The calculated heat contents of LaB_6 using the Neumann-Kopp rule show a significant deviation from the experimental data, which indicates that the present model is more reasonable.

Figure 4 presents the calculated activities of B at 1627 °C along with the experimental data.[2] The activities of La and B were both measured by Storms and Mueller[2] using vapor pressure method. Although these reported experimental data were not used in this optimization, the computed values are in consistent with the activity of B. Such a good agreement further verifies the robustness of the present modeling.

5.2 The La-Bi System

Figure 5 shows the calculated La-Bi phase diagram using the presently assessed parameters (solid line for set 1 and dashed line for set 2), compared with the experimental data,[14] where a good agreement is obtained. Table 4 shows the comparison between the measured and calculated invariant equilibria in the La-Bi binary system. It shows that most of the computed invariant temperatures agree well with the experimental data. The formation temperatures for La_4Bi_3 and LaBi are quite uncertain according to the literature.[14] During the assessment, it was found that when these two calculated temperatures are changed within the reported temperature range, the peritectic reaction is changed into eutectic one when using the substitutional model. In this assessment the reaction temperatures were expensed to ensure the right reaction type. However, when the liquid phase was described using the associate model (dashed line), both the calculated temperatures and reaction types for these two reactions agree reasonably with the measured data. It indicates that the associate model has advantages to handle such a situation.

Figure 6 shows the calculated enthalpies of formation at 298.15 K using two sets of parameters, compared with the experimental data,[20] first-principles calculations and the estimated values.[23,24] The differences between the calculated values via CALPHAD approach, first-principles calculations, estimated values and experimental data are within the experimental error.

6. Conclusion

- A critical assessment has been made for the experimental phase diagram and thermodynamic data in the La-B and La-Bi systems. The enthalpies of formation for the binary compounds in these two binary systems were obtained via the first-principles method.
- The La-B and La-Bi systems are modeled by means of the CALPHAD approach. The substitutional solution model and associate model are used to describe the liquid phase in the La-B and La-Bi systems. The calculated results are in good agreement with the experimental data for both systems.
- For the binary system with the nonsymmetrical liquidus, an associate model can describe the reaction type and temperatures more accurately than the substitutional solution model.

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